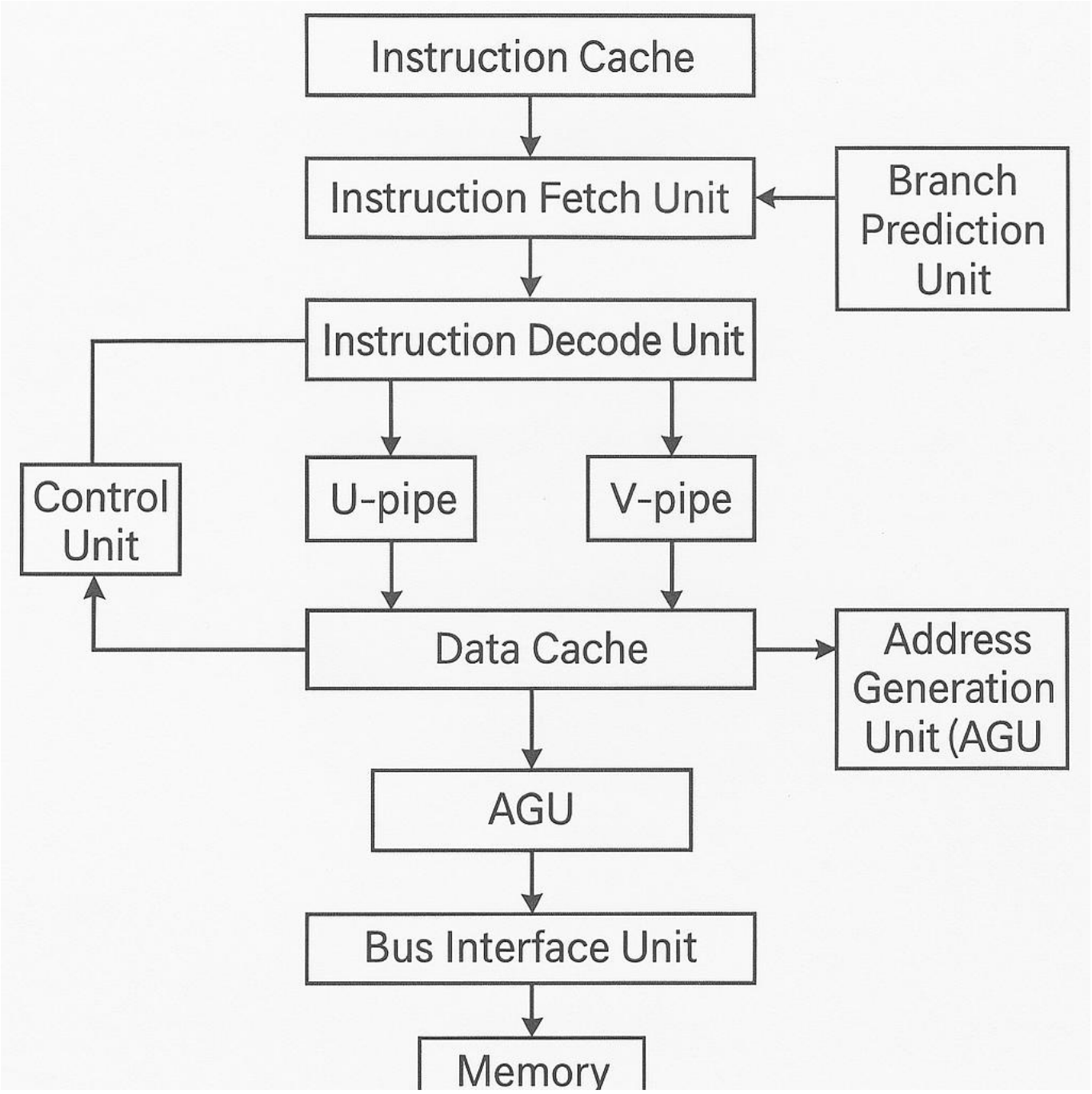
Instruction Cache
Instruction Fetch Unit
Branch Prediction Unit
Instruction Decode Unit
Control Unit
U-pipe
V-pipe
Data Cache
Address Generation Unit (AGU
AGU
Bus Interface Unit
Memory

```
GUI Turbo Assembler x64
≡ File  Edit  View  Run  Breakpoints  Data  Options  Window  Help                READY
Module: flag File: flag.asm 16                                                    1
  .MODEL SMALL
  .STACK 100H
  .DATA
      MSG DB
      HEX_CH
      FLAGS

  .CODE
  MAIN PROC
      MOV AX
      MOV DS

      PUSHF
      POP FL

      MOV DX
      MOV AH
      INT 21
Watches
[■]=CPU 80486                                                          3-[↑][↓]
#flag#main                                                   ax 0880    c=0
  cs:0000 B88008        ◆ MOV AX, DGROUP                     bx 0000    z=0
  cs:0003 8ED8          ◆ MOV DS, AX                         cx 0000    s=0
  cs:0005 9C            ◆ PUSHF ; Push flag regi             dx 0000    o=0
  cs:0006 8F062000      ◆ POP FLAGS ; Pop flag r             si 0000    p=0
  cs:000A BA0000        ◆ MOV DX, OFFSET MSG                 di 0000    a=0
  cs:000D B409          ◆ MOV AH, 09H                        bp 0000    i=1
  cs:000F CD21          ◆ INT 21H ; Print "Flag              sp 0100    d=0
  cs:0011 A12000        ◆ MOV AX, FLAGS ; Load f             ds 0880
  cs:0014 E80400        ◆ CALL PRINT_HEX ; Print             es 086C
  cs:0017 B44C          ◆ MOV AH, 4CH                        ss 0883
  cs:0019 CD21          ◆ INT 21H ; Exit program             cs 087C
#flag#print_hex                                              ip 000A
  es:0000 CD 20 7D 9D 00 EA FF FF = }¥ Ω
  es:0008 AD DE 32 0B C5 05 6B 07 ¡ Z╛╢♠k•
  es:0010 15 03 28 08 15 03 93 01 §♥(◘§♥ô⊖                    ss:0102 0403
  es:0018 01 01 01 00 02 04 FF FF ⊖⊖⊖ ◘♦                      ss:0100 52FB
F1-Help F2-Bkpt F3-Mod F4-Here F5-Zoom F6-Next F7-Trace F8-Step F9-Run F10-Menu
```